

Basic Array / Array Basic .java

marks = [0, 0, 0, 0, 0] ← int
names = [null, null, null] ← string
marks.length = 5

Main array :

a = [10, 20, 30, 40, 0, 0, 0, 0]
size = 4

shift: 1. Random Access

a[2] = 30 → a[2] = 30

a[4] =

2. Insert 15 at index 1

Before a = [10, 20, 30, 40, 0, 0, 0, 0]
size = 4 (value)

shift

a[4] = a[3] → [10, 20, 30, 40, 40, 0, 0, 0]

a[3] = a[2] → [10, 20, 30, 30, 40, 0, 0, 0]

a[2] = a[1] → [10, 20, 30, 40, 0, 0, 0]

Insert =

a[1] = 15

[10, 15, 20, 30, 40, 0, 0, 0]

size = 5

shift = 3

3. Insert 50 at end

$a[5] = 50$

Before $a = [10, 15, 20, 30, 40, 50, 0, 0]$

size = 6

shift = 0

4. Delete element at index 0

$a = [15, 20, 30, 40, 50, 0, 0, 0]$

size = 5

shift = 5 impractical

5. Array Bounds checks

$a[8] = 99$ Valid $i \rightarrow 0, 1, 2, 3, 4, 5, 6, 7$

$\uparrow$
error (exception)

$\rightarrow$ ArrayIndexOutOfBoundsException

6. Jagged 2-D Array

$\text{triangle} = \text{new int}[4][5];$

Initially:

$\text{triangle} = [\text{null}, \text{null}, \text{null}, \text{null}]$

$r = 0$ $\text{triangle}[0] = [1] \leftarrow$

$r = 1$: $+ [1] = [1, 1] \leftarrow$

$r = 2$: $+ [2] = [1, 2, 1] \leftarrow$

$r = 3$: $+ [3] = [1, 3, 3, 1] \leftarrow$

Final output
triangle

7. Grow Array

Before: $[15, 20, 30, 40, 50, 0, 0, 0]$
 $n = 8$

Arrays copy of C ($a, a.length \times 2$)

New length:
 $8 \times 2 = 16$

bigger = $[15, 20, 30, 40, 50, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0]$

final array (set start before)

$[15, 20, 30, 40, 50, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0]$

$\rightarrow [1] = [0]$ increment

$\rightarrow [1, 1] = [1] +$

$\rightarrow [1, 2, 1] = [2] +$

$\rightarrow [1, 2, 2, 1] = [0] +$